

Optimizing a Minimal Language using Pre-trained Language Models

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Agenda

Motivation and Concepts

Building the Language Step-by-Step

Results and Insights

Motivation and Concepts

Why Study Language Optimality?

- Human language evolved for survival and cooperation—not efficiency.
- Natural languages are shaped by history, noise, and cognitive constraints.
- Can we do better if we start from scratch?
- Constructed languages (ConLangs) offer a testing ground.

Trade-offs in Language Design

- Language design faces a trade-off between:
 - **Expressiveness**
 - **Efficiency**
- Natural languages aren't optimized for either.
- Our goal: can we define "optimal" and get closer to it?

Constructed Languages (ConLangs)

- ConLangs are intentionally built from first principles.
- Vary in purpose: auxiliary, fictional, logical, minimalist, expressive.
- We explore how to optimize a ConLang using modern tools.

Enter Language Models

- Large Language Models (LLMs) embed knowledge of real languages.
- Can they help design new ones?
- We use LLMs to:
 - Generate and translate language
 - Analyze simplicity, meaning retention, and structure

Thesis Goals

1. Define linguistic optimality along measurable axes.
2. Design a modular pipeline to create efficient ConLangs.
3. Use LLMs to guide generation, translation, and evaluation.

Building the Language Step-by-Step

The Journey Begins

- We start with a blank slate.
- Step-by-step, we add layers: phonetics → grammar → vocabulary → translation → evaluation.
- Each choice is a design decision.

Step 1: Choosing Phonemes

- First, define how words sound: phonemic inventory.
- Use the most common phonemes from PHOIBLE.
- Simpler inventories → easier learning.

Step 2: Structuring Sounds (Phonotactics)

- Rules for combining sounds.
- We test:
 - Basic CV (Consonant-Vowel) structures.
 - Custom syllable constraints.

Step 3: Choosing a Grammar

- What kind of grammar should we use?
- Try agglutinative (suffix-based) and isolating (word-order-based).
- Implement with reusable grammatical features.

Step 4: Building a Vocabulary

- Vocabulary = heart of expressiveness.
- We test multiple strategies:
 - From source
 - Simplified by clustering
 - Approximation via embeddings
- Trade-off: size vs. meaning retention.

Step 5: Translating Text

- Use LLMs to translate between English and our ConLang.
- Translate and back-translate to test loss of meaning.
- Input: paragraphs from known corpora.

Step 6: Evaluating Performance

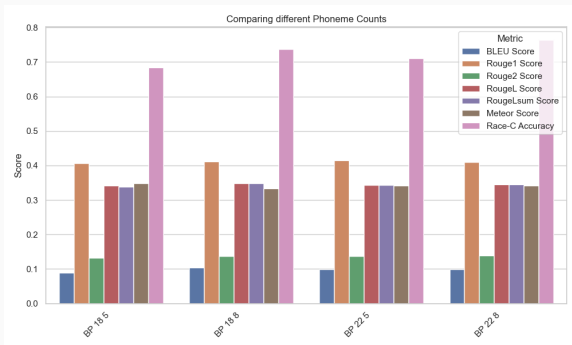
- We measure:
 - Information loss (BLEU, ROUGE, comprehension)
 - Simplicity (perplexity, vocabulary size, phoneme count)
 - Zipf's Law compliance
- Every pipeline produces scores for comparison.

- Modular codebase with JSON outputs.
- Local + remote LLMs.
- Embeddings via MiniLM.
- Clustering methods: Agglomerative, KMeans, HDBScan.

Results and Insights

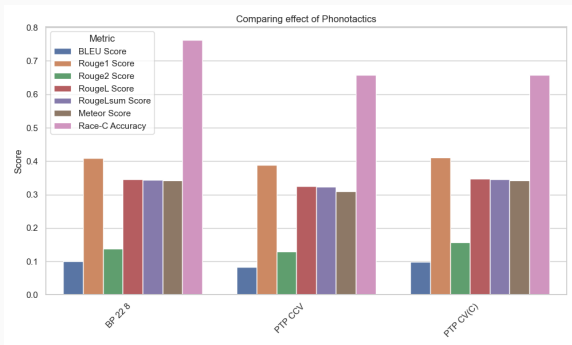
Does Phoneme Count Matter?

- Reducing phoneme count had minimal effect.
- Simpler phonology still retained meaning.



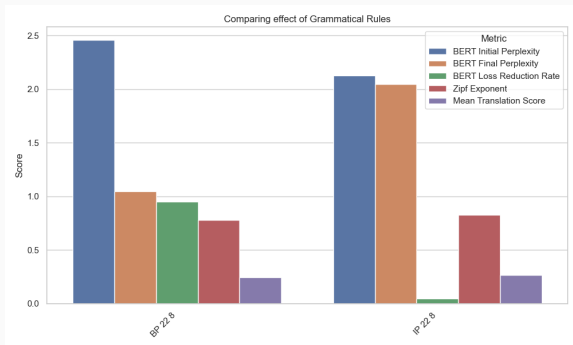
Effect of Phonotactics

- Simplified phonotactics had little impact on comprehension.
- Supports minimalist phonological design.



Grammar: Agglutinative vs Isolating

- Translation and comprehension scores were similar.
- Only language model perplexity showed differences.



- Simplified vocab reduced n-gram translation accuracy.
- Meaning-based (Race-C) scores dropped less.
- Better score-per-word ratios in simpler methods.

Vocabulary Performance Visuals

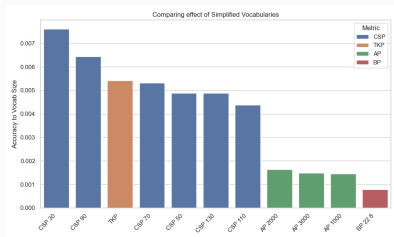


Figure 1: Race-C Accuracy

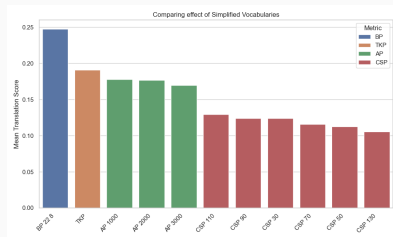
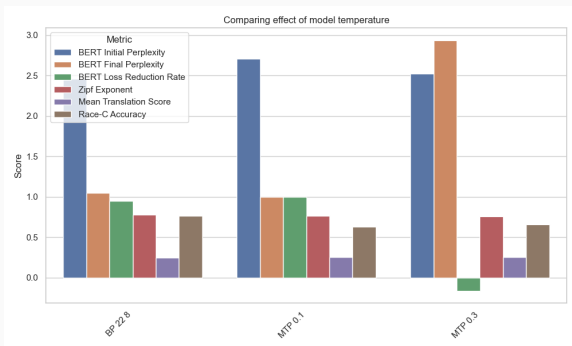


Figure 2: MTS Score

Language Model Temperature

- Higher temperature increases randomness.
- Lower values (0.2) more reproducible and stable.



Comparing Vocabulary Pipelines

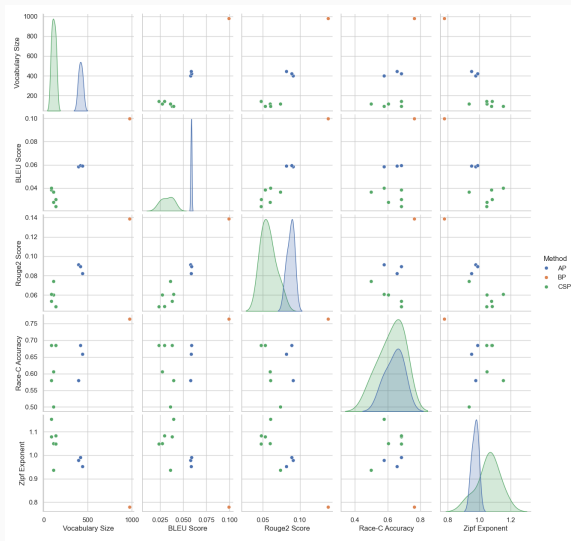


Figure 3: Comparison of different vocabulary generation strategies.

Conclusion

- LLMs can assist in building efficient ConLangs.
- Simpler phonology and grammar still retain meaning.
- Vocabulary trade-offs are the most sensitive.
- Modular pipeline allows repeatable experiments.

What's Next?

- Improve vocab generation using semantics.
- Evaluate spoken/multimodal ConLangs.
- Design user-facing ConLang creators.
- Explore LLM co-evolution with constructed languages.

Questions?

Thank you!
Questions are welcome.